SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Concept Mapping (5 QUESTIONS)

Course Code:	ITA0510	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 2: Concept Mapping
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	I. Sarathy	**Reg. No.:**	192412599
Section / Batch:	B/2024	**Date:**	9/10/26

Code of Conduct

I Sarathy I(192412599) *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Sarathy .I

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Concept Mapping

Q1. An automobile company is developing an autonomous vehicle that uses cameras to detect vehicles, pedestrians, and traffic signs. Objects closer to the camera appear larger, while distant objects appear smaller. **Prepare a concept map applying the image formation and camera projection model to show how a three-dimensional road scene is converted into a two-dimensional image. Include the relationships between focal length, object distance, image distance, and projected image size.**

Q2. A manufacturing company uses cameras to measure the dimensions of bottles moving on a conveyor belt. Engineers observe that bottles at different distances from the camera appear to have different sizes. **Prepare a concept map applying perspective projection to show how focal length, camera position, viewing angle, object distance, and image size are related. Use the concept map to represent how these factors influence the accuracy of bottle measurement.**

Q3. A construction company uses a drone camera to capture images of buildings from different heights and viewing angles. The appearance and measured dimensions of the building change when the drone position is changed. **Prepare a concept map applying image formation and projection concepts to show the relationship between camera orientation, focal length, perspective projection, camera position, calibration, and the accuracy of building measurements.**

Q4. An industrial company wants to select a suitable projection model for road monitoring and inspection of objects placed on a flat conveyor belt. The engineers are considering perspective and orthographic projection for these applications. **Prepare a concept map comparing perspective and orthographic projection based on depth, object size, distance, parallel lines, and image formation. Use the concept map to identify the appropriate projection model for each application.**

Q5. A robotics company wants a robot to identify the position and shape of objects placed on a table using a camera. The camera captures only a two-dimensional representation of the three-dimensional objects. **Prepare a concept map applying the image formation model to show the relationship between a 3D object, camera parameters, projection geometry, focal length, camera orientation, and the resulting 2D image. Show how these concepts help determine the object's position and appearance.**

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ____________ Date: 9/10/26	Signature: ____________ Date: ____________	Signature: ____________ Date: ____________

[illegible] Computer vision

[illegible] AI [illegible] Concept mapping

Image formation and camera projection model

3D Real Scene

Visible, illuminations and light signal

↓

Camera and Image formation

Light reflected from object enters the camera lens

↓

Perspective projection

3D coordinates are projected onto a 2D image plane

↓

2D Image

objects appear at different sizes and positions

Focal length (f)

Larger f gives greater image magnification for fixed object distance

Object distance (z)

Greater distance produces a smaller image

Image distance (v)

Distance from the lens to the image plane

Projected size (h)

Depends on object size, focal length and distance

2) Perspective projection for Bottle measurement

Bottle on conveyor belt

↓

Camera parameters

Focal length + camera position + viewing angle

↓

Perspective projection

[illegible] size depends on object

↓

object distance
Increasing distance reduces projected size
Viewing angle
Can introduce foreshortening and shape distortion

focal length
affect image magnification
Camera position
determines the viewing perspective and depth

3) Image-Based Building measurement using Camera parameters and projection.

3D Building
Actual height, width and depth
↓
Place Camera position and orientation
Rotation, Translation, pitch, yaw and roll
↓
Camera parameters
focal length, principal point and lens
↓
perspective projection
3D building points become 2D image points
↓
Camera calibration and geometric Correction
↓
Accurate Building measurement

Comparison of perspective and orthographic projection

3D Scene
Camera projects 3D object into a 2D image
↓

erspective projection
g distance affects
ject size

orthographic projection
= projection rays are
parallel

(x,y)   [illegible]/101)

*Apparent size decreases with depth	* Object size does not change with depth
* parallel lines may converage at vanishing	* parallel lines remain in the image
* Depth affects image Scale	* Depth does not affect projected Scale

5) Image Formation model for Robot Object Identification:

3D Object on a Table
Actual position, dimensions and Shape

↓

Image Formation
Reflected light Passes through the camera lens

Camera Intrinsics   Camera Extrinsics

↓

Project Geometry
Maps 3D Coordinates to 2D pixel

↓

2D image processing
Object detection, segmentation and feature extraction

↓

Robot position and Shape Estimation
Estimate object location & dimensions